

FABIO CENTOFANTI

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Section of Statistics and Data Science

Department of Mathematics

KU Leuven

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EXPERIENCE

Postdoctoral researcher

Statistics and Data Science Section, Department of Mathematics, KU Leuven, Belgium

1 January 2025–ongoing

Postdoctoral researcher (Ricercatore TD-A, SECS/S-02)

Department of Industrial Engineering, University of Naples Federico II, Naples, Italy

11 September 2023–31 December 2024

Postdoctoral researcher (Assegno di ricerca)

Department of Industrial Engineering, University of Naples Federico II, Naples, Italy

Interpretable Statistical Methods for Monitoring Energy Consumption in Industrial Processes Based on the Analysis of Complex Data

1 December 2022–10 September 2023

Postdoctoral researcher (Assegno di ricerca)

Department of Civil, Building, and Environmental Engineering, University of Naples Federico II, Naples, Italy

Application of Functional Clustering and Regression Techniques to the Urban Traffic Context

1 February 2022–31 October 2022

Postdoctoral researcher (Borsa di ricerca)

Department of Electrical Engineering and Information Technology, University of Naples Federico II, Naples, Italy

Interpretable methods for functional data analysis and prediction – Metodi statistici interpretabili per l'analisi e la previsione di dati funzionali

15 September 2021–15 January 2022

EDUCATION

University of Naples Federico II, Naples, Italy

Ph.D., XXXIII Cycle, Industrial Engineering

17 September 2021, *Optimum cum laude*

- Thesis title: *Statistical Methodologies of Functional Data Analysis for Industrial Applications.*
- Supervisors: Prof. Biagio Palumbo, Ph.D. and Prof. Simone Vantini, Ph.D.

University of Naples Federico II, Naples, Italy

M.Sc., Management Engineering

October 2017, *110/110 cum laude*

- Thesis title: *A Functional Data Analysis Approach to Real-Time Monitoring of Ship Operational Conditions and Prediction of CO₂ Emissions.*
- Advisors: Prof. Biagio Palumbo, Ph.D. and Antonio Lepore, Ph.D.

University of Naples Federico II, Naples, Italy

B.Sc., Management Engineering

October 2015, *110/110 cum laude*

Music Conservatory San Pietro a Majella, Naples, Italy

M.A., Principal Piano

March 2015, *10/10 cum laude and special mention*

Science High School Leon Battista Alberti, Naples, Italy

Secondary School Diploma
July 2012, 100/100

VISITING

Politecnico di Milano, Milan, Italy

MOX, Laboratory for Modeling and Scientific Computing, Department of Mathematics “Francesco Brioschi”
January 2018–July 2018

Technical University of Denmark, Copenhagen, Denmark

Department of Applied Mathematics and Computer Science
January 2021–April 2021

Katholieke Universiteit Leuven (KU Leuven), Leuven, Belgium

Department of Mathematics

- February 2023–July 2023

- 17 March 2024–25 March 2024

- 13 September 2024–15 November 2024

ARTICLES IN SCIENTIFIC JOURNALS

1. Palumbo, B., Centofanti, F., and Del Re, F. (2020). Function-on-function regression for assessing production quality in industrial manufacturing. *Quality and Reliability Engineering International*, 36(8), 2738–2753.
2. Centofanti, F., Lepore, A., Menafoglio, A., Palumbo, B., and Vantini, S. (2021). Functional regression control chart. *Technometrics*, 63(3), 281–294.
3. Capezza, C., Centofanti, F., Lepore, A., Menafoglio, A., Palumbo, B., and Vantini, S. (2021). Functional regression control chart for monitoring ship CO₂ emissions. *Quality and Reliability Engineering International*, 38(3), 1519–1537.
4. Capezza, C., Centofanti, F., Lepore, A., and Palumbo, B. (2021). Functional clustering methods for resistance spot welding process data in the automotive industry. *Applied Stochastic Models in Business and Industry*, 37(5), 908–925.
5. Capezza, C., Centofanti, F., Lepore, A., and Palumbo, B. (2021). A functional data analysis approach for the monitoring of ship CO₂ emissions. *Gestão & Produção*, 28(3).
6. Centofanti, F., Fontana, M., Lepore, A., and Vantini, S. (2022). Smooth lasso estimator for the function-on-function linear regression model. *Computational Statistics & Data Analysis*, 176, 107556.
7. Centofanti, F., Lepore, A., Menafoglio, A., Palumbo, B., and Vantini, S. (2023). Adaptive smoothing spline estimator for the function-on-function linear regression model. *Computational Statistics*, 38(1), 191–216.
8. Capezza, C., Centofanti, F., Lepore, A., Menafoglio, A., Palumbo, B., and Vantini, S. (2023). Funcharts: Control charts for multivariate functional data in R. *Journal of Quality Technology*, 55(5), 566–583.
9. Centofanti, F., Colosimo, B. M., Grasso, M. L., Menafoglio, A., Palumbo, B., and Vantini, S. (2023). Robust functional ANOVA with application to additive manufacturing. *Journal of the Royal Statistical Society Series C: Applied Statistics*, 72(5), 1210–1234.
10. Centofanti, F., Lepore, A., and Palumbo, B. (2024). Sparse and smooth functional data clustering. *Statistical Papers*, 65(2), 795–825.
11. Capezza, C., Centofanti, F., Lepore, A., and Palumbo, B. (2024). Robust Multivariate Functional Control Chart. *Technometrics*, 66(4), 531–547.
12. Capezza, C., Capizzi, G., Centofanti, F., Lepore, A., and Palumbo, B. (2024). An adaptive multivariate functional EWMA control chart. *Journal of Quality Technology*, 57(1), 1–15.
13. Centofanti, F., Lepore, A., Kulahci, M., and Spooner, M. P. (2025). Real-time monitoring of functional data. *Journal of Quality Technology*, 57(2), 135–152.
14. Centofanti, F., Lepore, A., and Palumbo, B. (2025). An Adaptive Multivariate Functional Control Chart. *Technometrics*, 67(4), 603–616.
15. Centofanti, F., Hubert, M., Palumbo, B., and Rousseeuw, P. J. (2025). Multivariate Singular

Spectrum Analysis by Robust Diagonalwise Low-Rank Approximation. *Journal of Computational and Graphical Statistics*, 34(1), 360–373.

16. Simonelli, F., Centofanti, F., Lepore, A., Pariota, L., Marzano, V., and Palumbo, B. (2025). Data-driven investigation of traffic spatial correlation through functional data analysis. *Transportation Research Part C: Emerging Technologies*, 174, 105088.
17. Capezza, C., Centofanti, F., Forcina, D., Lepore, A., and Palumbo, B. (2026). Functional Mixture Regression Control Chart. *The Annals of Applied Statistics*, 20(1), 683–701.
18. Centofanti, F. (2026). Statistical Process Monitoring Based on Functional Data Analysis. *WIREs Computational Statistics*, 18(1).
19. Hirari, M., Centofanti, F., Hubert, M., and Van Aelst, S. (2026). Casewise and Cellwise Robust Multilinear Principal Component Analysis. *Journal of Computational and Graphical Statistics*.
20. Centofanti, F., Hubert, M., and Rousseeuw, P. J. (2026). Robust Principal Components by Case-wise and Cellwise Weighting. *Technometrics*.

PREPRINTS

1. Centofanti, F., Hubert, M., and Rousseeuw, P. J. (2025). Cellwise and Casewise Robust Covariance in High Dimensions. *arXiv preprint arXiv:2505.19925*.
2. Hirari, M., Centofanti, F., Hubert, M., and Van Aelst, S. (2026). Robust Tensor-on-Tensor Regression. *arXiv preprint arXiv:2603.25911*.

BOOK CHAPTERS

1. Colosimo, B. M., and Centofanti, F. (2022). Model Interpretability, Explainability and Trust for Manufacturing 4.0. In *Interpretability for Industry 4.0: Statistical and Machine Learning Approaches*, pp. 21–36.
2. Capezza, C., Centofanti, F., Lepore, A., Menafoglio, A., Palumbo, B., and Vantini, S. (2023). Functional Regression Control Charts with an Application to Ship Fuel Consumption Monitoring. *Wiley StatsRef: Statistics Reference Online*, 1–10.
3. Capezza, C., Centofanti, F., Lepore, A., Menafoglio, A., Palumbo, B., and Vantini, S. (2024). Statistical Process Monitoring Methods Based on Functional Data Analysis. In *Multimodal and Tensor Data Analytics for Industrial Systems Improvement*, Springer Optimization and Its Applications, vol. 211, pp. 41–55. Springer.
4. Capezza, C., Centofanti, F., Lepore, A., and Palumbo, B. (2014). Robust Statistical Monitoring of a Resistance Spot Welding Process. *Wiley StatsRef: Statistics Reference Online*, 1–8.

CONFERENCE CONTRIBUTIONS AND TALKS

1. Bocchetti, D., Lepore, A., Palumbo, B., **Capezza, C.**, and Centofanti, F. Normalized ship fuel-speed curve for monitoring CO₂ emissions by multi-sensor system data. *Abstract at the 17th Annual Conference of the European Network for Business and Industrial Statistics*, Naples, Italy, 9–14 September 2017.
2. Bocchetti, D., Capezza, C., Centofanti, F., Lepore, A., Di Matteo, R., Palumbo, B., and **Vitiello, L.** A statistical framework for monitoring, reporting and verification of CO₂ emissions in the maritime transport. *Abstract at the 17th Annual Conference of the European Network for Business and Industrial Statistics*, Naples, Italy, 9–14 September 2017.
3. **Centofanti, F.**, Lepore, A., Menafoglio, A., Palumbo, B., and Vantini, S. A functional regression control chart for profile monitoring. *Proceeding at the 49th Scientific Meeting of the Italian Statistical Society*, Palermo, Italy, 20–22 June 2018.
4. Centofanti, F., **Lepore, A.**, Menafoglio, A., Palumbo, B., and Vantini, S. Run-length performance estimation of the functional regression control chart. *Abstract at the Ninth Workshop on Simulation*, Barcelona, Spain, 25–29 June 2018 (invited).
5. Centofanti, F., **Lepore, A.**, Menafoglio, A., Palumbo, B., and Vantini, S. Functional regression control chart for CO₂ emission monitoring. *Abstract at the 18th Annual Conference of the European Network for Business and Industrial Statistics*, Nancy, France, 2–5 September 2018 (invited).
6. **Centofanti, F.**, Fontana, M., Lepore, A., and Vantini, S. Sparse estimation of function-on-function

- non-concurrent linear regression models. *Abstract at the Final CRoNoS Events and Workshop on Multivariate Data Analysis 2019*, Limassol, Cyprus, 14–16 April 2019 (invited).
7. **Centofanti, F.**, Fontana, M., Lepore, A., and Vantini, S. A sparse estimator for the function-on-function linear regression model. *Proceeding at SIS 2019 – Smart Statistics for Smart Applications*, Milan, Italy, 18–21 June 2019 (invited).
 8. Centofanti, F., **Lepore, A.**, Menafoglio, A., Palumbo, B., and Vantini, S. Functional regression control chart. *Abstract at the 6th International Symposium on Statistical Process Monitoring*, Tunis, Tunisia, 8–12 July 2019 (invited).
 9. Palumbo, B., **Del Re, F.**, and Centofanti, F. Statistical engineering approach to additive manufacturing via functional data analysis. *Abstract at the 19th Annual Conference of the European Network for Business and Industrial Statistics*, Budapest, Hungary, 2–4 September 2019.
 10. **Centofanti, F.**, Fontana, M. An interpretable estimator for the function-on-function linear regression model with application to the Canadian weather data. *Proceeding at the 50th Scientific Meeting of the Italian Statistical Society*, 2020 (invited).
 11. **Centofanti, F.**, Lepore, A., Menafoglio, A., Palumbo, B., and Vantini, S. Functional regression control chart with application to ship CO₂ emission monitoring. *Abstract at the Virtual 2020 INFORMS Annual Meeting*, 8–13 November 2020 (invited).
 12. **Centofanti, F.**, Lepore, A., Menafoglio, A., Palumbo, B., and Vantini, S. A spatially adaptive estimator for the function-on-function linear regression model with application to the Swedish Mortality dataset. *Proceeding at the 50th Scientific Meeting of the Italian Statistical Society*, 21–25 June 2021.
 13. **Centofanti, F.** Robust functional ANOVA with application to additive manufacturing. *ENBIS Workshop: Interpretability for Industry 4.0*, Naples, Italy, 12–13 July 2021 (invited).
 14. **Centofanti, F.**, Lepore, A., and Palumbo, B. Sparse and smooth cluster analysis of functional data. *Abstract at the ENBIS-21 online conference*, 13–15 September 2021 (invited).
 15. **Centofanti, F.**, Lepore, A., Kulahci, M., and Spooner, M. P. Real-time monitoring of functional data. *Abstract at the ENBIS-21 online conference*, 13–15 September 2021.
 16. **Capezza, C.**, Centofanti, F., Lepore, A., and Palumbo, B. Analysis of resistance of spot welding process data in the automotive industry via functional clustering techniques. *Abstract at the ENBIS-21 online conference*, 13–15 September 2021.
 17. **Centofanti, F.**, Lepore, A., and Palumbo, B. Sparse and smooth clustering for functional data. *Abstract at the 2021 INFORMS Annual Meeting*, 24–27 October 2021 (invited).
 18. **Capezza, C.**, Centofanti, F., Lepore, A., Menafoglio, A., Palumbo, B., and Vantini, S. funcharts: An R package for the real-time monitoring of multivariate functional data. *Abstract at the 14th International Conference of the ERCIM*, London, UK, 18–20 December 2021 (invited).
 19. **Centofanti, F.**, Colosimo, B. M., Grasso, M. L., Menafoglio, A., Palumbo, B., and Vantini, S. Robust analysis of variance for functional data. *Abstract at the 2022 ISBIS Conference*, Napoli, Italy, 20–21 June 2022 (invited).
 20. Capezza, C., Centofanti, F., **Lepore, A.**, and Palumbo, B. Functional clustering techniques applied to resistance spot welding curves in the automotive industry 4.0. *Abstract at the 2022 ISBIS Conference*, Napoli, Italy, 20–21 June 2022 (invited).
 21. **Centofanti, F.**, Colosimo, B. M., Grasso, M. L., Menafoglio, A., Palumbo, B., and Vantini, S. A robust approach for functional ANOVA with application to additive manufacturing. *Proceeding at the 51th Scientific Meeting of the Italian Statistical Society*, Caserta, Italy, 22–24 June 2022.
 22. Capezza, C., Centofanti, F., **Lepore, A.**, and Palumbo, B. Interpretability in functional clustering with an application to resistance spot welding process in the automotive industry. *Proceeding at the 51th Scientific Meeting of the Italian Statistical Society*, Caserta, Italy, 22–24 June 2022.
 23. **Capezza, C.**, Centofanti, F., Lepore, A., Menafoglio, A., Palumbo, B., and Vantini, S. An R package for the statistical process monitoring of functional data. *Proceeding at the 51th Scientific Meeting of the Italian Statistical Society*, Caserta, Italy, 22–24 June 2022.
 24. **Centofanti, F.**, Colosimo, B. M., Grasso, M. L., Menafoglio, A., Palumbo, B., and Vantini, S. Functional analysis of variance in presence of outliers: the RoFANOVA approach. *Abstract at the 2022 ENBIS Conference*, Trondheim, Norway, 26–30 June 2022.
 25. **Capezza, C.**, Centofanti, F., Lepore, A., Menafoglio, A., Palumbo, B., and Vantini, S. Statistical

- process control of multivariate functional data in R. *Abstract at the 2022 ENBIS Conference*, Trondheim, Norway, 26–30 June 2022.
26. **Centofanti, F.**, Lepore, A., and Palumbo, B. Sparse clustering of functional data. *Abstract at the 2022 European Conference on Data Analysis*, Naples, Italy, 14–16 September 2022.
 27. **Capezza, C.**, Centofanti, F., Lepore, A., and Palumbo, B. Robust Control Charts for Multivariate Functional Data. *Abstract at the 15th International Conference of the ERCIM*, London, UK, 17–19 December 2022 (invited).
 28. Capezza, C., **Centofanti, F.**, Lepore, A., and Palumbo, B. Robust Profile Monitoring for Multivariate Functional Data. *Abstract at ICORS2023: International Conference on Robust Statistics*, Tolosa, France, 22–26 May 2023 (invited).
 29. **Centofanti, F.**, Lepore, A., and Palumbo, B. Sparse clustering of functional data. *Proceeding at CLADAG 2023 14th Scientific Meeting*, Salerno, Italy, 11–13 September 2023 (invited).
 30. **Centofanti, F.**, Lepore, A., and Palumbo, B. Sparse and smooth clustering of functional data. *Abstract at the 16th International Conference of the ERCIM WG on Computational and Methodological Statistics*, Berlin, Germany, 13–15 December 2023 (invited).
 31. Centofanti, F., **Capezza, C.**, Forcina, D., Lepore, A., and Palumbo, B. Control charts for functional data based on functional mixture regression. *Abstract at the 16th International Conference of the ERCIM WG on Computational and Methodological Statistics*, Berlin, Germany, 13–15 December 2023 (invited).
 32. **Centofanti, F.**, Hubert, M., Palumbo, B., and Rousseeuw, P. J. RODESSA for robust multivariate singular spectrum analysis. *Proceeding at the 52nd Scientific Meeting of the Italian Statistical Society*, Bari, Italy, 17–20 June 2024.
 33. **Centofanti, F.**, Hubert, M., Palumbo, B., and Rousseeuw, P. J. Robust Multivariate Singular Spectrum Analysis by RODESSA. *Abstract at the 2nd INFORMS Conference on Quality, Statistics and Reliability*, Como, Italy, 1–4 July 2024.
 34. **Centofanti, F.**, Hubert, M., Palumbo, B., and Rousseeuw, P. J. Multivariate singular spectrum analysis by robust diagonalwise low-rank approximation. *Abstract at ICORS2024: International Conference on Robust Statistics*, Washington D.C., USA, 29 July–1 August 2024 (invited).
 35. **Centofanti, F.**, Hubert, M., Palumbo, B., and Rousseeuw, P. J. Multivariate Singular Spectrum Analysis by Robust Diagonalwise Low-Rank Approximation. *Abstract at the 24th Annual Conference of the European Network for Business and Industrial Statistics*, Leuven, Belgium, 15–19 September 2024.
 36. **Centofanti, F.**, Hubert, M., and Rousseeuw, P. J. Cellwise and Casewise Robust Covariance in High Dimensions. *Abstract at the International Conference on Robust Statistics 2025*, Stresa, Italy, 19–23 May 2025 (invited).
 37. **Centofanti, F.** Robust Principal Components by Casewise and Cellwise Weighting. *Contributed talk at the Leuven Statistics Days*, KU Leuven, Leuven, Belgium, 26–28 May 2025.
 38. **Centofanti, F.**, Hubert, M., and Rousseeuw, P. J. Cellwise and Casewise Robust Covariance in High Dimensions. *Abstract at the 15th Scientific Meeting of CLADAG*, University of Naples Federico II, Naples, Italy, 8–10 September 2025 (invited).
 39. **Centofanti, F.**, Lepore, A., and Palumbo, B. An Adaptive Multivariate Functional Control Chart. *Abstract at the 25th Annual Conference of the European Network for Business and Industrial Statistics*, University of Piraeus, Piraeus, Greece, 14–18 September 2025 (invited).
 40. **Centofanti, F.**, Hubert, M., and Rousseeuw, P. J. Cellwise and Casewise Robust Multivariate Regression and Inference. *Abstract at the IMS Annual Meeting 2026*, Salzburg Congress, Salzburg, Austria, 6–9 July 2026 (invited).
 41. **Centofanti, F.**, Dağdır, C. H., Hubert, M., and Rousseeuw, P. J. Robust classification and regression with casewise and cellwise outliers. *Abstract at the 19th Conference of the International Federation of Classification Societies*, Milan, Italy, 14–16 July 2026 (invited).
 42. **Centofanti, F.**, Hubert, M., and Rousseeuw, P. J. Cellwise and Casewise Robust Multivariate Regression and Inference. *Abstract at the International Conference on Robust Statistics*, Istanbul, Türkiye, 20–24 July 2026 (invited).

INVITED SEMINARS

1. Sparse and smooth functional data clustering. *The fourth One World webinar organized by YoungStatS: Recent Advances in Functional Data Analysis*, 30 June 2021.
2. Robust Diagonalwise Multivariate Singular Spectrum Analysis. *Research Institute for Statistics and Information Science*, University of Geneva, Geneva, Switzerland, 27 October 2023.
3. Multivariate Singular Spectrum Analysis by Robust Diagonalwise Low-Rank Approximation. *Department of Mathematics*, KU Leuven, Leuven, Belgium, 31 October 2024.
4. Cellwise and Casewise Robust Covariance in High Dimensions. *Department of Economics and Law*, University of Cassino, Cassino, Italy, 12 September 2025.

BOOKS

- Palumbo, B., Lepore, A., Capezza, C., and Centofanti, F. (2018). *Introduzione ai modelli di analisi multivariata a variabili latenti. Applicazioni per l'ingegneria*. Aracne editrice.

MEMBERSHIP IN SOCIETIES

- SIS Italian Statistical Society, 2018–today.
- ySIS – Young SIS, 2018–today.
- European Network for Business and Industrial Statistics (ENBIS), 2018–today.
- Institute for Operations Research and the Management Sciences (INFORMS), Quality, Statistics, and Reliability (QSR) section, 2022–today.
- Classification and Data Analysis Group (CLADAG), 2026–today.

AWARDS, GRANTS, AND HONOURS

Personal Awards

- Winner of the *ENBIS Young Statistician Award 2026*, awarded by the European Network for Business and Industrial Statistics (ENBIS) to young statisticians for significant contributions to the field.
- Winner of the *Chikio Hayashi Award 2026*, awarded by the International Federation of Classification Societies (IFCS) to young researchers for contributions to classification, clustering, and related areas.
- Winner of the *Young Researcher Paper Award* at the CLADAG Conference 2023, 14th Scientific Meeting, awarded by the Classification and Data Analysis Group (CLADAG).
- Selected among the *best graduate students* of the Bachelor's degree in Management Engineering of the year 2015, Department of Industrial Engineering, University of Naples Federico II, April 2016.

Grants

- *ENBIS knowledge fund* for participation at the ENBIS 2024 conference, 500€ + free registration at the conference.
- *Grant for a scientific stay in Flanders 2024*, 4,356.00€, Research Foundation – Flanders (FWO).
- *Grant for a scientific stay in Flanders 2023*, 5,082.00€, Research Foundation – Flanders (FWO).
- Program of *Scambi Internazionali per mobilità di breve durata*, 1,949.50€, University of Naples Federico II.

Honours

- Member elected of the ySIS board for 2022–2023.
- Member of the Local Organising Committee of the 2022 ISBIS Conference, Napoli, Italy, 20–21 June 2022.
- Member of the Program Committee of the Convegno Intermedio SIS, Ancona, Italy, 21–23 June 2023.
- National Scientific Qualification (ASN) as Associate Professor in Statistics, obtained in 2025.
- Member of the Local Organising Committee of the *Second Edition of the Rousseeuw Prize for Statistics*, KU Leuven, 2024.
- Member of the Local Organising Committee of the *Third Edition of the Rousseeuw Prize for*

SOFTWARE PACKAGES

- Capezza, C., Centofanti, F., Lepore, A., Menafoglio, A., Palumbo, B., and Vantini, S. (2023). **funcharts**: Functional Control Charts. R package version 1.6.0.
- Centofanti, F. (2026). **slasso**: S-LASSO Estimator for the Function-on-Function Linear Regression. R package version 1.0.1.
- Centofanti, F. (2026). **rofanova**: Robust Functional Analysis of Variance. R package version 1.0.1.
- Centofanti, F., Lepore, A., Menafoglio, A., Palumbo, B., and Vantini, S. (2023). **adass**: Adaptive Smoothing Spline (AdaSS) Estimator for the Function-on-Function Linear Regression. R package version 1.0.0.
- Centofanti, F. (2026). **sasfunclost**: Sparse and Smooth Functional Clustering. R package version 1.0.1.

REFEREEING ACTIVITY

Associate Editor

Statistics and Computing, 2026–ongoing.

Referee for international peer-reviewed journals, including *Journal of Quality Technology*, *Quality Technology & Quantitative Management*, *Statistical Papers*, *Computational Statistics*, *Statistics and Computing*, *Computational Statistics and Data Analysis*, *Statistical Analysis and Data Mining*, *Journal of Computational and Graphical Statistics*, and *Technometrics*.

TEACHING AND MENTORING ACTIVITIES

Supervision of Master's Degree Theses

- Orsola Spiezia (2017). *A statistical approach for the monitoring of a complex system for modern trains heating, ventilation and air conditioning system*. Management Engineering, University of Naples Federico II.
- Francesca D'Ambrosio (2017). *Data exploration and performance control of a complex system for modern rail vehicles heating, ventilation and air conditioning application*. Management Engineering, University of Naples Federico II.
- Vincenzo Napolitano (2022). *Adaptive multivariate functional EWMA control chart for resistance spot welding profiles*. Mechanical Engineering, University of Naples Federico II.
- Davide Forcina (2023). *Functional mixture regression control chart for resistance spot welding profiles*. Mechanical Engineering, University of Naples Federico II.
- Arthur Daisomont (2025). *Casewise, cellwise, and blockwise robust PCA method*. Mathematics, KU Leuven.
- Anna Zhuk (2026). *Robust functional principal component analysis*. Mathematics, KU Leuven.
- Flavian Dumélié (2026). *Casewise and cellwise robust partial least squares*. Mathematics, KU Leuven.

Teaching Activities

- 2017–2018. Teaching Assistant, *Statistical Quality Control*, University of Naples Federico II.
- 2018–2019. Teaching Assistant, *Statistics for Technology*, University of Naples Federico II.
- 2018–2019. Teaching Assistant, *Statistical Methods for Industrial Process Monitoring Control*, University of Naples Federico II.
- 2019–2020. Teaching Assistant, *Statistical Methods for Industrial Process Monitoring*, University of Naples Federico II.
- 2021–2022. Lecturer within the PhD program in Industrial Engineering, University of Naples Federico II. Course: *Statistical Process Monitoring of Complex Engineering Data*.
- 2025. Supervision of bachelor projects in statistics, KU Leuven.
- 2025–ongoing. Responsible for the project assessment component, Master's course *Robust Statistics* (Mathematics), KU Leuven.

OTHER CONTRIBUTIONS AND INITIATIVES

- Centofanti, F., Lepore, A., Menafoglio, A., Palumbo, B., and Vantini, S. Functional Regression Control Chart: a New Framework for Profile Monitoring. *Contribution to YoungStats: the blog of Young Statisticians Europe (YSE)*, 2020.
- Organizer of the *Seminar Series on Robust Statistics and Data Science*, Section of Statistics and Data Science, Department of Mathematics, KU Leuven, 2025–ongoing.

May 2, 2026